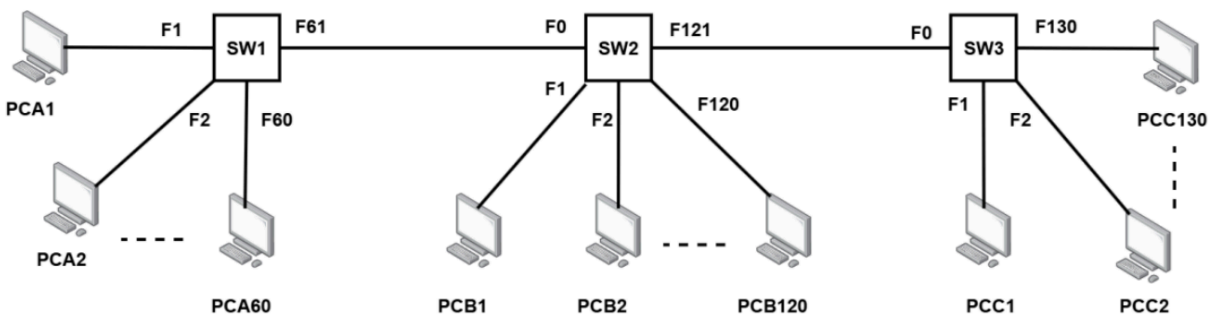
Lecture 15 - Data Link Layer

Summer 2025 (A)

6.



Consider the above switched network. The host PCB1, connected to SW2 sends a frame to PCC2, connected to SW3. All switch tables are empty except SW3, which has the information of PCC2 in its table.

I. How many PCs will receive the request frame? Why?

II. Draw the switch tables of all 3 switches once the reply is received by PCB1.

I. How many PCs will receive the request frame? Why?

Answer: 181 PCs will receive the request frame.

The Reasoning:

1. **Source:** PCB1 sends a frame to PCC2. Since the switch tables for SW1 and SW2 are empty, they do not know the location of the destination (PCC2) and must perform **flooding** (unicast flooding).
2. **SW2 Action:** When PCB1 sends the frame into SW2 (via port F1), SW2 floods it out of all other active ports:
   - To all other PCs directly connected to it: PCB2 through PCB120 (**119 PCs**).
   - Out of port F0 to **SW1**.
   - Out of port F121 to **SW3**.
3. **SW1 Action:** SW1 receives the flooded frame from SW2. Since its table is empty, it floods the frame to all its connected PCs: PCA1 through PCA60 (**60 PCs**).

4. **SW3 Action:** SW3 receives the frame from SW2. According to the prompt, **SW3 already has the information for PCC2 in its table**. Therefore, SW3 does *not* flood the frame. It performs a **selective forward** directly to PCC2 (**1 PC**).
5. **Total Calculation:** 119 (on SW2) + 60 (on SW1) + 1 (PCC2 on SW3) = 181 PCs.

II. Switch Tables after the Reply is Received by PCB1

When PCC2 sends a reply back to PCB1, the switches learn the locations of both devices. A switch table typically maps a **MAC Address** (Host) to a specific **Interface** (Port).

SW1 Table

SW1 learned about PCB1 when the initial request was flooded from SW2.

Host (MAC Address)	**Port (Interface)**
PCB1	F61

SW2 Table

SW2 learned about PCB1 from the initial request and learned about PCC2 when the reply came back from SW3.

Host (MAC Address)	**Port (Interface)**

PCB1	F1
PCC2	F121

SW3 Table

SW3 already knew PCC2. It learned about PCB1 when the initial request arrived from SW2.

Host (MAC)	Port
PCB1	F0
PCC2	F2 (or F130)*

11.A PC (**192.168.10.50**) prints to its network printer (**192.168.10.200, MAC: 0C-43-00-12-BC-DD**). The printer is replaced with a new one using the same IP address. Initially, the PC cannot print, but after approximately 10 minutes, printing works automatically.
I. **Why** could not the PC print immediately after the printer was replaced?
II. **What** could have been done to enable printing immediately after the printer replacement?

Scenario:

- **PC IP:** 192.168.10.50
- **Old printer IP:** 192.168.10.200, MAC: 0C-43-00-12-BC-DD
- **New printer:** same IP (192.168.10.200), different MAC

I. Why could the PC not print immediately?

- PCs use **ARP (Address Resolution Protocol)** to map IP addresses to MAC addresses.
- The PC had cached the **old printer's MAC address** for IP 192.168.10.200 in its **ARP table**.
- When the printer was replaced, the **MAC changed**, but the PC still tried to send packets to the **old MAC address**, so printing failed.
- After 10 minutes, the **ARP cache timed out**, and the PC sent a new ARP request → learned the new printer's MAC → printing worked.

Key point: The delay was due to the **ARP cache holding the old MAC address**.

II. How to enable printing immediately?

To fix this instantly:

1. **Clear the PC's ARP cache** manually:
    - On Windows: `arp -d 192.168.10.200`
    - On Linux/macOS: `sudo ip -s -s neigh flush 192.168.10.200`
2. **Ping the printer** to force the PC to update the ARP entry:
    - `ping 192.168.10.200` → PC learns new MAC
3. Alternatively, **restart the PC or network interface**, which also clears the ARP cache.

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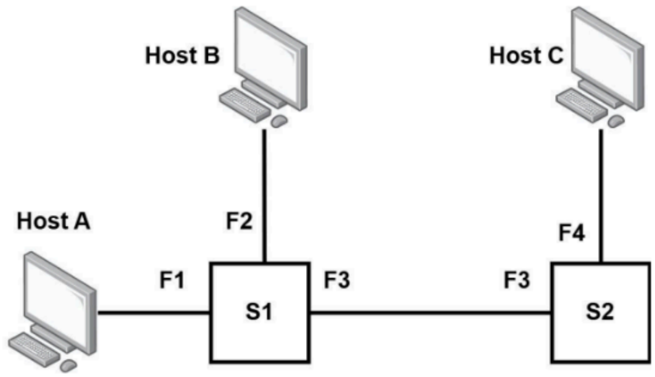
11.

Host A (IP: 192.168.1.10, MAC: AA-AA-AA-AA-AA-AA), Host B (IP: 192.168.1.12, MAC: BB-BB-BB-BB-BB-BB) and Host C (IP: 192.168.1.20, MAC: CC-CC-CC-CC-CC-CC) are connected according to the topology on the side. Initially, all ARP and MAC tables are empty. Host A wants to send a message to Host C.



I. What kind of ARP packet (mention source and destination MAC address) will Host A generate, and how will the switches process it?

II. Switches are termed “plug-and-play”. Briefly explain the significance.

Part I: ARP Packet Generation and Switch Processing

Since Host A needs to send a message to Host C (IP: 192.168.1.20) but does not know Host C's physical address, it must first use the **Address Resolution Protocol (ARP)**.

1. The ARP Packet Details

Host A will generate an **ARP Request** packet. Because Host A doesn't know who owns the target IP, this packet must be sent to everyone on the local network.

- **Type:** ARP Request (Broadcast)
- **Source MAC:** AA-AA-AA-AA-AA-AA (Host A)
- **Destination MAC:** FF-FF-FF-FF-FF-FF (The Layer 2 Broadcast address)
- **Target IP inside payload:** 192.168.1.20

2. How the Switches Process It

Since the MAC tables are initially empty, the switches will perform two main actions: **Learning** and **Flooding**.

- **Switch S1:**
  1. **Learning:** S1 receives the frame on port **F1**. it records in its MAC table that AA-AA-AA-AA-AA-AA is reachable via F1.
  2. **Flooding:** Because the destination is a broadcast address (FF-FF-FF-FF-FF-FF), S1 forwards the packet out of all other active ports: **F2** (to Host B) and **F3** (toward S2).
- **Switch S2:**
  1. **Learning:** S2 receives the frame on its port **F3**. It records that AA-AA-AA-AA-AA-AA is reachable via its F3 port.
  2. **Flooding:** Like S1, it recognizes the broadcast address and forwards the packet out of all other ports: **F4** (to Host C).

Outcome: Host B receives the packet but drops it because the IP doesn't match. Host C receives it, recognizes its own IP, and prepares an ARP Reply.

Part II: Significance of "Plug-and-Play"

Switches are described as **plug-and-play** (or self-learning) because they require zero manual configuration to start forwarding traffic efficiently between devices.

Key Aspects:

- **Self-Learning:** As shown in Part I, switches automatically build their **MAC Address Tables** by observing the source MAC addresses of incoming frames. They don't need an administrator to tell them which computer is on which port.
- **Automatic Maintenance:** The switch dynamically updates its table. If you unplug Host A and move it to a different port, the switch "re-learns" the new location as soon as Host A sends its next frame.
- **Transparency:** To the end hosts (Host A, B, and C), the switches are "transparent." The hosts send data as if they are directly connected, and the switches handle the delivery logic in the background.

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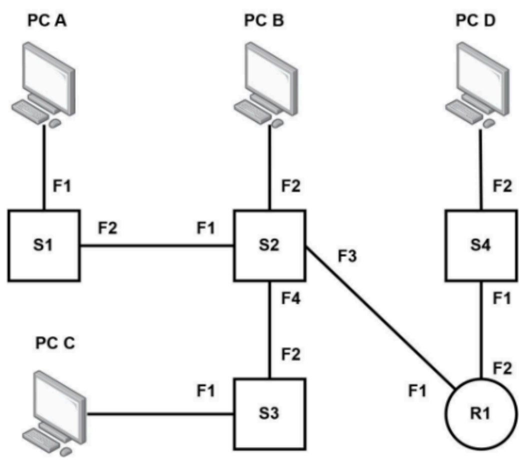

Refer to the figure, **PC A** sends an ARP requests for **PC B**.

I. **State** the **source** and **destination** MAC addresses in the ARP request packet.

II. **Explain** how PC B knows that it has to reply.

III. **State** what will router R1 do with the packet and **why**.

I. Source and Destination MAC Addresses

In an **ARP (Address Resolution Protocol) Request**, the goal is to find a MAC address corresponding to a known IP address. Because PC A does not yet know where PC B is, it must send the packet to everyone on the local network.

- **Source MAC Address:** The MAC address of **PC A** (e.g., MAC_A).
- **Destination MAC Address:** The **Broadcast Address**, which is FF:FF:FF:FF:FF:FF.

II. How PC B knows it has to reply

When a device receives an ARP request, it looks at the **Target IP Address** field inside the ARP message (the data portion of the packet).

1. The ARP request is a broadcast, so every device in the local segment (PC B, PC C, and Router R1) receives and processes the frame.
2. Each device compares its own IP address with the **Target IP Address** specified in the request.
3. **PC B** will see that the requested IP address matches its own.

4. Consequently, PC B will process the request and send back an **ARP Reply** containing its own MAC address. Other devices (like PC C) will see the mismatch and silently discard the packet.

III. What Router R1 will do and why

Action: Router R1 will **receive the packet but will not forward it** to the network segment containing PC D.

Reason: * **Broadcast Boundaries:** ARP requests are Layer 2 broadcasts. Routers, by design, do not forward Layer 2 broadcast frames to other interfaces. They define the boundary of a "Broadcast Domain."

- **Local Scope:** Since ARP is used to resolve addresses on the local link, R1 assumes the target is within its own local network segment. If R1 were to forward broadcasts, it would cause unnecessary traffic (and potentially broadcast storms) across the entire internetwork.

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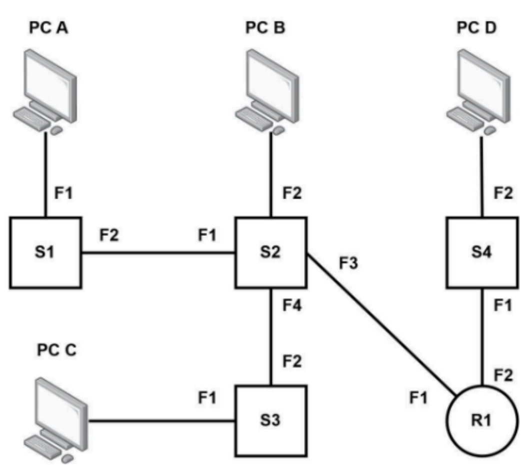

Refer to the figure given, **PC B** sends a packet to **PC C**, at this stage all switch tables contain information about all devices shown.

I. **State** the actions that the switch **S1** will take after it receives that packet.

II. **Explain**, using the above scenario, why we call switches to be transparent.

I. Actions of Switch S1

The prompt states that **PC B** sends a packet to **PC C** and that **all switch tables contain information about all devices shown**.

1. **Packet Path Identification**: PC B is connected to S2. To reach PC C (which is connected to S3), the logical path is through S2 to S3.
2. **S1's Reception**: Since S1 is connected to S2 via port F2, it will receive the packet if S2 broadcasts it or if there is specific traffic intended for a device on S1's segment. However, in this specific scenario (PC B to PC C), S1 will likely receive the frame if S2 treats it as a broadcast or if the network is a shared medium.
3. **The "Filter" Action**: Since the switch table is already fully populated, S1 looks up the **Destination MAC address** (PC C).
4. **Action**: S1 will see that PC C is **not** located on its port F1 (where PC A is). Because the destination is not on any of its other active ports leading to the destination, and it already knows where all devices are, **Switch S1 will filter (drop) the packet**. It realizes the packet does not need to traverse S1 to reach its destination.

II. Why Switches are "Transparent"

In networking, a switch is called **transparent** because the end devices (PC B and PC C) are completely unaware of its existence.

- **No Configuration Needed on Host**: PC B does not need to know the MAC address of S2, S3, or S1 to send data to PC C. It only needs the destination MAC of PC C.
- **Packet Integrity**: The switch does not modify the source or destination IP/MAC addresses of the Ethernet frame as it passes through. From the perspective of PC B, it is talking "directly" to PC C.
- **Invisible Operation**: The switches build their forwarding tables (MAC tables) automatically by observing traffic. They "disappear" into the infrastructure, acting

as a bridge that connects segments without requiring the end-users to manage the hop-by-hop path.

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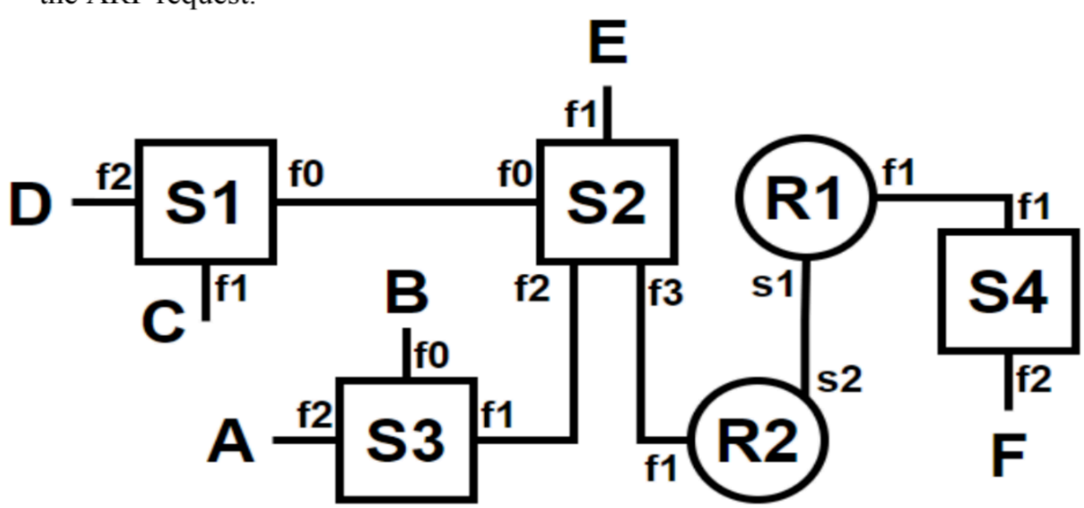
Refer to the figure given below.

I. Device **E** sends an ARP request for the MAC address of device **A**. **State** which intermediary devices will receive the ARP request frame and forward it to other ports. Also **state** which intermediary devices will drop the frame.

II. **Show** the entries of the MAC address tables of S2 and S3 after receiving the reply to the ARP request.



I. ARP Request Propagation

An ARP request is sent as a **Broadcast frame** (destination MAC: FF:FF:FF:FF:FF:FF). This means every switch that receives it will flood the frame out of all ports except the one it arrived on. Routers, however, do not forward broadcast traffic by default.

1. Intermediary devices that receive and forward the request:

- **S2:** Receives the frame from E on port f1. It floods the frame out of ports f0, f2, and f3.
- **S1:** Receives the broadcast from S2 on port f0. It floods it out of ports f1 and f2.

- **S3:** Receives the broadcast from S2 on port f1. It floods it out of ports f0 and f2.
- **R2:** Receives the broadcast from S2 on port f1. However, as a router, it processes the packet but does **not** forward the broadcast frame to other interfaces (like s2).

2. Intermediary devices that drop the frame:

- **R2:** While R2 receives the frame, it "drops" it in terms of forwarding; it will not pass the broadcast to R1.
- **R1 and S4:** These devices will **never receive** the ARP request because R2 acts as a boundary for broadcast traffic.

II. MAC Address Tables for S2 and S3

When Device A sends the **ARP Reply** back to E, it is a **Unicast frame**. As this reply travels from A to E, the switches (S3 and S2) learn the MAC addresses and ports for both the source (A) and the destination (E).

Assuming the MAC address of Device A is MAC_A and Device E is MAC_E:

MAC Address Table of S3

Port	MAC Address
f2	MAC_A (Learned from the source of the ARP reply)
f1	MAC_E (Learned from the initial ARP request broadcast)

MAC Address Table of S2

Port	**MAC Address**
f2	MAC_A (Learned when the ARP reply arrives from S3)
f1	MAC_E (Learned from the initial ARP request source)

10.Given your MAC address is AF:CC:FE:12:23:40.

I. **Identify** the OUI part of the MAC address.

II. **Discuss** why the MAC address is considered to be a flat address.

Given MAC address: AF:CC:FE:12:23:40

I. OUI (Organizationally Unique Identifier) part

- A MAC address is **48 bits (6 bytes)**:
  - First **3 bytes (24 bits)** → **OUI** (assigned to manufacturer)
  - Last **3 bytes (24 bits)** → device-specific NIC part
- For AF:CC:FE:12:23:40:
  - **OUI (first 3 bytes):** AF:CC:FE
  - **Device-specific part (last 3 bytes):** 12:23:40

Answer: OUI = AF:CC:FE

II. Why a MAC address is considered a flat address

1. **No hierarchy:**
    - Unlike IP addresses, MAC addresses **do not encode network location**.
    - They simply identify a device **uniquely** across all networks.
2. **Flat structure:**
    - The OUI + NIC part ensures global uniqueness.
    - There is **no subnetting or routing information** in the MAC.
3. **Implication:**
    - Routers cannot determine network topology from a MAC.
    - MAC addresses are primarily used **within LANs** for local delivery.

11.A router gets an ARP request packet. **List the steps** that the router will perform in all possible scenarios. **State** what is in the destination MAC address of the ARP packet that the router received.

When a router receives an ARP (Address Resolution Protocol) request, its behavior depends on whether the request is meant for the router itself, a device on another interface, or if it's simply a broadcast it should ignore.

The Destination MAC Address

The destination MAC address of an incoming ARP request packet is always the **Broadcast Address**: FF:FF:FF:FF:FF:FF.

This is because the sender knows the target's IP address but does not know its MAC address, so it must broadcast the query to every device on the local network segment.

Scenario 1: The Router is the Intended Target

In this case, the "Target Protocol Address" (IP) in the ARP packet matches the IP address assigned to the router's interface that received the packet.

1. **Receive and Unpack:** The router receives the broadcast frame and passes the ARP packet to its processor.
2. **Verify IP:** The router checks the Target IP field. It sees that the IP matches its own.
3. **Update ARP Cache:** The router typically records the Sender's IP and MAC address in its own ARP table for future use.
4. **Generate ARP Reply:** The router creates an ARP Reply packet.
5. **Unicast Response:** It sends the reply directly back to the sender (Unicast), placing its own hardware MAC address in the "Sender Hardware Address" field.

Scenario 2: The Router Acts as a Proxy (Proxy ARP)

If Proxy ARP is enabled, a router may respond to an ARP request for an IP address that is **not** its own, provided the router knows how to reach that destination.

1. **Receive and Verify:** The router sees the Target IP is not its own.
2. **Check Routing Table:** The router checks if it has a specific route to that Target IP via a different interface.
3. **Assess "Reachability":** If the router determines it is the best path to that IP, it acts as a "proxy."
4. **Send Proxy Reply:** The router sends an ARP Reply back to the original sender, providing **its own MAC address** as the destination for that remote IP.

Scenario 3: The Target is Another Device (Default Behavior)

This is the most common scenario for standard traffic where the router is not the target and Proxy ARP is disabled.

1. **Receive and Verify:** The router processes the broadcast and checks the Target IP.
2. **Mismatch Found:** The router determines the Target IP does not match its interface IP.

3. **Discard:** Since the router is not the owner of the IP and is not configured to act as a proxy, it **drops (discards) the packet**. It does not forward ARP requests to other subnets, as ARP is a Layer 2 broadcast protocol and is confined to the local broadcast domain.

Summary Table: Router Actions

Scenario	Target IP Matches Router?	Action Taken
Direct Target	Yes	Sends ARP Reply with its MAC address.
Proxy ARP	No (but route exists)	Sends ARP Reply with its MAC address.
Normal Traffic	No	Discards the packet.

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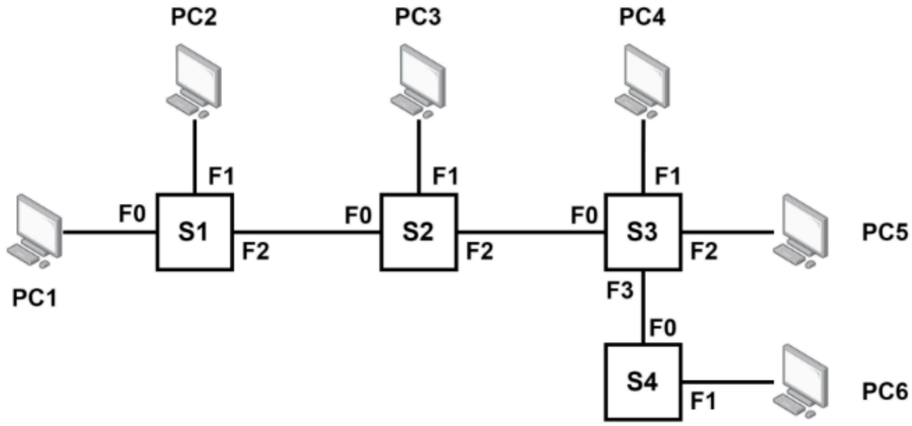

Suppose, the initial MAC Address tables of all switches are empty. PC2 sends two frames, one to PC4 and the other to PC6. How does switch S1 forward these two frames? Consider that PC4 does not reply, but PC6 does. Now, **illustrate** the current state of the MAC Address tables of switches S3 and S4. [Mention the device name and the interface]

1. How Switch S1 Forwards the Frames

When **PC2** sends frames to **PC4** and **PC6**:

- **Step 1:** The frames arrive at Switch **S1** on interface **F1**.
- **Learning:** S1 records that the MAC address of PC2 is reachable via its F1 port.
- **Forwarding:** Since the MAC address tables are initially empty, S1 does not know where PC4 or PC6 are located. Therefore, S1 **floods** the frames out of all its other active interfaces (**F0** and **F2**).

2. State of MAC Address Tables (S3 and S4)

We must track the sequence of events to see what each switch "sees" on its ports.

Sequence of Events:

1. **PC2 sends frames:** The frames travel from S1 -> S2 -> S3.
   - S3 receives the frame from S2 on its **F0** port. It learns PC2 is at F0.

    - S3 floods the frame to PC4 (F1), PC5 (F2), and S4 (F3).
    - S4 receives the frame from S3 on its **F0** port. It learns PC2 is at F0.
2. **PC4 does not reply:** No new information is learned about PC4.
3. **PC6 replies:** A frame travels from PC6 -> S4 -> S3.
    - S4 receives the reply on **F1**. It learns PC6 is at F1.
    - S3 receives the reply from S4 on **F3**. It learns PC6 is at F3.

MAC Address Table: Switch S3

MAC Address (Device)	Interface
PC2	F0
PC6	F3

Note: S3 does not know PC4's location because PC4 never sent a frame.

MAC Address Table: Switch S4

MAC Address (Device)	Interface
PC2	F0
PC6	F1

10. Why does any entry of an ARP table have a TTL? ARP requests are never forwarded by routers - **Explain why.**

I. Why does an ARP table entry have a TTL?

An **ARP table entry** has a **Time To Live (TTL)** to ensure correctness and efficiency:

- **IP-MAC mappings can change** due to device reboot, NIC replacement, or mobility.
- TTL **prevents stale entries** from remaining in the table.
- It forces **periodic re-learning** of valid MAC addresses.
- Helps **manage memory** by removing unused entries automatically.

Without TTL, incorrect MAC mappings could persist and cause packet loss.

II. Why are ARP requests never forwarded by routers?

- **ARP operates at Layer 2** and uses **broadcast frames** (FF:FF:FF:FF:FF:FF).
- **Routers separate broadcast domains** and **do not forward Layer-2 broadcasts**.
- ARP is meant to resolve MAC addresses **only within the local network**.
- For remote destinations, a host ARPs **only for the default gateway's MAC**, not the remote host.

Forwarding ARP would cause broadcast flooding and scalability problems, so routers block it.

12.Given your MAC address is **AF:CC:FE:12:23:40**.

I. Identify if the address is locally administered or globally unique

II. Discuss why the MAC address is considered to be a portable address

I. Locally administered or globally unique?

- Look at the **first byte**: AF → 10101111 (binary)
- **U/L bit** = second least significant bit of first byte → **1**
  - 0 → Globally unique (vendor-assigned)
  - 1 → Locally administered

Answer: Locally administered (U/L bit = 1)

II. Why is a MAC address considered portable?

- **Independent of network location** – it stays the same when a device moves between networks.
- **Operates at Layer 2** – not tied to IP or routing.
- **Can be manually set (locally administered)** – useful for virtualization, privacy, and mobility.

Answer: MAC addresses are **portable** because they uniquely identify a device at the Data Link Layer, independent of the network it is connected to.